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The relation, $\omega = \frac{v}{r}$ could be used.

Angular momentum, $L = mr^2 \times \frac{v}{r}$

$= mvr$

$$= 9.1 \times 10^{-31} \times 2.2 \times 10^{6} \times 0.53 \times 10^{-10}$$

$$L = 1.06 \times 10^{-34}\ kg\ m^2\ s^{-1}$$

3. A solid sphere of mass 20 kg and radius 0.25 m rotates about an axis passing through the center. What is the angular momentum if the angular velocity is 5 rad s^{-1}

Solution:

Mass of the sphere, m = 20 kg
Radius r = 0.25 m
Angular velocity $\omega = 5$ rad s^{-1}

Solution:

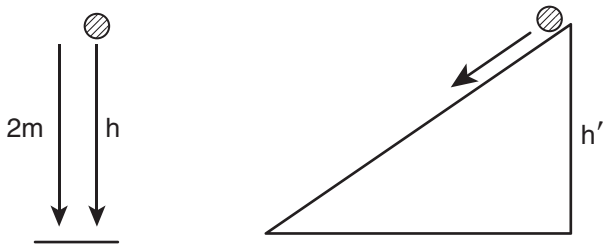
Angular momentum $L = I\omega = \frac{2}{5} mr^2\omega$

$$= \frac{2}{5} \times 20 \times (0.25)^2 \times 5 = 40 \times (0.0625) = 2.5$$

$$L = 2.5 \text{ kg m}^2\text{s}^{-1}$$

4. A solid cylinder when dropped from a height of 2 m acquires a velocity while reaching the ground. If the same cylinder is rolled down from the top of an inclined plane to reach the ground with same velocity, what must be the height of the inclined plane? Also compute the velocity.

Solution:



In the first case,

potential energy = kinetic energy

$$mgh = \frac{1}{2} mv^2$$

$$mg \times 2 = \frac{1}{2} mv^2 \qquad (1)$$

In second case,

potential energy = translational kinetic energy + rotational kinetic energy

$$mgh' = \frac{1}{2} mv^2 + \frac{1}{2} I\omega^2$$

$$mgh' = \frac{1}{2} mv^2 + \frac{1}{2}\left(\frac{mr^2}{2}\right)\left(\frac{v^2}{r^2}\right)$$

$$\therefore mgh' = \frac{3}{4} mv^2 \qquad (2)$$

Dividing (2) by (1),

$$\frac{mgh'}{mg \times 2} = \frac{\frac{3}{4} mv^2}{\frac{1}{2} mv^2} = \frac{3}{4} \times \frac{2}{1} = \frac{3}{2}$$

$$h' = 3m$$

From equation (1), $2mg = \frac{1}{2} mv^2$

$$v = \sqrt{4g} = 2\sqrt{g}$$

$$v = 2 \times \sqrt{9.81}$$

$$v = 6.3\, m\, s^{-1}$$